

Lesson plan

Name of the Programme:	B.Tech(Hons.)	
Title of the Course:	Digital Systems & Computer Architecture	
Course code	CS1101	
Semester:	I	
Names of the Course Instructor(s)	Dr. Vidya M J, Prof. Aisiri K S ,Dr. R.V. Madhusudhan,Dr. Shyam Mohan, Prof. Shilpa Hiremath, Dr. Nagaraj J	
Course credit	No. of hours per week (2+0+2)	Total no. of Teaching hours
03	04 Hours	60 Hours

Session	Module No.	Topic	Pedagogy /activities	Pre-reading/reference
1.	1	SI units; charge, current, voltage; energy & power; safety	Chalk talk +quick numericals	Floyd/Buchla Ch.1–2
2.		Resistors; Ohm's Law; resistor color codes; power ratings	Worked examples	Floyd/Buchla Ch.2
3.		Construct Resistive Circuit and Verify Equivalent Resistance (Req) using KCL, KVL and Ohm's Law	Laboratory: To construct series and parallel resistive circuits using KCL, KVL and Ohm's Law	External Resources
4.				
5.		Series & parallel resistors; equivalent resistance techniques	Quick numericals +Worked examples	Floyd/Buchla Ch.3
6.		Sensors overview; independent voltage/current sources	Quick numericals +Worked examples	Floyd/Buchla Ch.3
7.		Construct Series and Parallel Resistive Circuit, calculate the Req, Voltage and Current for the different values of resistors and verify the result using a multimeter.	Laboratory: To construct a series and parallel resistive circuit using and Calculate the Requ, Voltage and Current.	External Resources
8.				

9.		Conservation of energy; KCL; KVL	Board derivations + short quiz	Floyd/Buchla Ch.3
10.		Capacitors & inductors: fundamentals; energy storage	Concept check	Floyd/Buchla Ch.4–5
11.		Construct a Resistor cube and calculate the total Resistance of Face, Edge and Body diagonals of the cube.	Laboratory: To construct a resistor cube and derive the resistance values of edge, face diagonal and body diagonal.	External Resources
12.				
13.		Conservation of energy; KCL; KVL; Passive filters (RC/RL) — qualitative; time constant intuition	Concept check+ Problem solving +Short numericals	Floyd/Buchla Ch.5
14.		Transistors (BJT) as switches; logic-level notion	Guided discussion	Boylestad (BJT/MOSFET)
15.		Construct Basic gates(AND, OR and NOT) using Transistors and verify the output using the truth table.	Laboratory: To construct basic logic gates such as NOT, AND and OR using transistors	External Resources
16.				
		Conductors, insulators, semiconductors (overview)	Concept map	Boylestad (Devices)
17.		Diodes: I-V, ideal vs practical; applications (clip/clamp intro)	Chalk talk + examples	Boylestad (Diodes)
		Unit-I consolidation: mixed numericals & concept recap	MCQ drill	Unit-I notes
18.		Construct Basic gates(AND, OR and NOT) using Diodes and verify the output using the truth table.	Laboratory: To construct basic logic gates such as NOT, AND and OR using Diodes.	External Resources
19.				
20.	2	Analog vs Digital; positional notation; number bases	Board drills	Mano & Ciletti Ch.1–2

21.	Conversions: bin $\leftrightarrow$ oct $\leftrightarrow$ hex $\leftrightarrow$ dec (methods & practice)	Quickfire exercises	Mano & Ciletti Ch.2
22.	Binary codes: weighted codes, BCD	Examples + short quiz	Mano & Ciletti Ch.2
23.	Construct Half-Adder using basic logic gates and verify the output using truth table.	Laboratory: To Construct half adder using basic logic gates.	External Resources
24.			
25.	Non-weighted codes: Gray, Excess-3; ASCII; 7-segment encoding	Decode/encode tasks	Mano & Ciletti Ch.2
26.	Basic logic gates; truth tables; universal gates (NAND/NOR)	Chalk talk +Quiz	Mano & Ciletti Ch.2–3
27.	Boolean algebra laws; DeMorgan; canonical forms (SOP/POS)	Worked examples	Mano & Ciletti Ch.3
28.	Construct half-subtractor using basic logic gates and verify the output using truth table	Laboratory: To Construct half subtractor using basic logic gates.	External Resources
29.			
30.	Simplification using Boolean algebra; typical patterns	Whiteboard practice	Mano & Ciletti Ch.3
31.	Karnaugh maps (2–4 variables); hazards (brief)	Pair problems	Mano & Ciletti Ch.3–4
32.	Combinational design: Half-Adder; Full-Adder concepts	Design walkthrough	Mano & Ciletti Ch.3
33.	Construct a Full adder circuit using 2 Half adders using universal gates and verify the output using the truth table.	Laboratory: To construct full adder using 2 half adders and construct basic logic gates using universal NAND and NOR gates.	External Resources
34.			
35.	Subtractor circuits: Half & Full; two's-complement subtract	Problem set	Mano & Ciletti Ch.3

36.		Data-routing blocks: Decoder, Encoder — use cases	Mini design tasks	Mano & Ciletti Ch.4
37.		MUX/DEMUX; function realization with MUX; Unit-II wrap	Integrative exercise	Mano & Ciletti Ch.4
38.	3	Build a square wave generator using the IC 555 Timer and observe the output using glowing LEDs.	Laboratory: To build a square wave generator using IC 555	External Resources
39.				
40.		Seq. vs comb.; clocking; classification; storage elements; SR-Latch; D-Latch; level vs edge triggering	Concept check+Timing sketches	Mano & Ciletti Ch.5
41.		Edge-triggered D-FF; timing parameters (setup/hold); metastability (intro)	Board examples	Mano & Ciletti Ch.5
42.		JK & T flip-flops; excitation tables; typical uses	Worked tables	Mano & Ciletti Ch.5
43.		Construct a SR-Latch using NAND and NOR gates and verify the output using the truth table.	Laboratory: Build S'R'-Latch using NOR, NAND gate and verify function table.	External Resources
44.				
45.		Registers; shift registers; serial $\leftrightarrow$ parallel conversion	Case examples	Mano & Ciletti Ch.5
46.		Counters: ripple vs synchronous; mod-N design (overview)	Design sketches	Mano & Ciletti Ch.5
47.		FSM fundamentals: Mealy vs Moore; state diagrams; design steps	Guided example	Mano & Ciletti Ch.6
48.		FSM examples: sequence detector; state reduction & assignment; Unit-III recap	Review Qs	Mano & Ciletti Ch.6



49.	4	Evolution of computers; Moore's law; performance & CPU clocks	Short lecture + Q&A	Harris & Harris Ch.1
50.		Sequential model; von Neumann model — components & flow	Diagram walk-through	Harris & Harris Ch.1
51.		Design of Arithmetic Logic Unit	Lab Activity: Understanding behaviour of arithmetic logic unit from working module.	External Resources
52.				
53.		Harvard architecture; comparison with von Neumann	Mini-debate	Harris & Harris Ch.1
54.		Architecture recap; simple case study tying concepts	Concept map	Harris & Harris Ch.1
55.	5	CPU design: datapath overview; GPRs; instruction formats (overview)	Block-diagram talk	Harris & Harris Ch.2–3
56.		Instruction cycle: fetch–decode–execute; control signals; control unit intro	Chalk talk + examples	Harris & Harris Ch.3
57.		CPU Design	Lab Activity: experiment is to show the basic top level functionality, organization and architecture of a computer	External Resources
58.				
59.		Memory hierarchy & types: registers, cache, main memory; locality intuition	Concept questions	Harris & Harris Ch.4
60.		Datapath & control unit integration; end-to-end flow; grand syllabus review	Oral Q&A + summary	Harris & Harris Ch.2–4

Assessment and Evaluation

Continuous Internal Evaluation (CIE): 70%	
Components of CIE	Weightage of each component
CIE-1 • Quiz • Assignments based on Tinker-Cad Software	10 marks out of 70 10 marks out of 70
CIE-2 • Theory	25 marks out of 70
CIE-3 • Lab record maintenance • Lab execution, verification of output & viva	10 marks out of 70 15 marks out of 70

Semester End Exams (SEE): 30%	
Mode of Exam	Theory
Weightage of Exam	30%

Course Outcomes- Programme Outcomes Matrix

Course Outcomes		PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	Understand the Fundamental Principles of Electronic Circuits	3	2	-	-	1	-	-	-	-	-	-	1
CO2	Apply and analyze the Concepts of Combinational and Sequential Circuits	3	3	2	1	1	-	-	-	-	-	-	1

CO3	Analyze the instruction cycle, execution cycle, data transfer processes, and overall functioning of the CPU within a computer system	3	2	-	-	-	-	-	-	-	-	-	1
CO4	Build Analog and Digital Logic Circuits using discrete components	3	3	2	1	1	-	-	-	2	2	1	1

Rubrics for evaluation of CIE (separate for each CIE component)

Total Marks		20
CIE-1 Components		Rubrics for Assessment
CIE-1	Component 1	The quiz will be conducted for 20 marks. Each right answer carries 0.5 mark. No negative marking for wrong answers. and the total marks is considered as average of 10 Marks
	Component-2	Assignments based on Tinker-Cad Software = 10 marks <i>(Refer Table-1)</i>

Table-1:

Graded Component-2 (10 Marks):

Grading Rubrics for Assignments based on Tinker-Cad Software (Total: 10 Marks)					
Sl. No	Criteria	Excellent	Very Good	Good	Needs Improvement
1.	Circuit Building(4 Marks)	Circuit fully functional, neat, and	Minor connection or layout	Works partially or with	Circuit incomplete or not working.

		accurate in all connections.	issues, overall works correctly.	noticeable errors.	
2.	Component List (2 Marks)	All components listed with correct names and specifications	Most components listed, minor detail missing.	Some components missing or mislabeled.	Component list incomplete or incorrect.
3.	Schematic(2Marks)	Clear, correct schematic with proper symbols and labeling.	Mostly correct with small symbol or wiring errors.	Understandable but contains multiple inaccuracies.	Unclear, incorrect, or missing schematic.
4.	Submission on Time(2 Marks)	Submitted on or before the deadline.	Submitted within the same day after deadline.	Submitted within 2 days after the deadline.	Submitted more than 2 days late

Total Marks		
CIE-2 Components-3		Rubrics for Assessment
CIE-2	Theory	Test will be conducted for 25 marks and will be evaluated according to Scheme of Evaluation prepared by team of course faculty.

Total Marks		25
CIE-3 Components		Rubrics for Assessment
CIE-3	Component - 4	Record maintenance=10 marks (<i>Refer Table-2</i>)
	Component - 5	Lab Exam+ Viva = 15 marks (<i>Refer Table-3</i>)

Table- 2

Graded Component-4(10 Marks)

Rubrics for Lab Write-up and Execution (Max: 10 marks)						
Sl. No.	Criteria	Measuring Method	Excellent	Good	Satisfactory	Poor
1	Datasheet write up (3 Marks)	Visual Aids (Circuit/ logic diagram, Graphs, Tables)	Visual aids are correctly represented, accurate, and well-integrated. (3 Marks)	Visual aids are mostly accurate, with minor issues (1.5 Mark)	Visual aids are basic, with some inaccuracies or integration issues. (1.0 Marks)	Visual aids are poorly integrated or often inaccurate. (0.5 Marks)
2	Circuit building with team collaboration (3 Marks)	Observation	Circuit is built accurately and efficiently, with excellent teamwork (3 Marks)	Correct calculations with minor errors. Proper use of graphs and tables. Results are interpreted correctly. (1.5 Mark)	Calculations are mostly correct but incomplete. Graphs and tables are used but may contain errors. (1.0 Marks)	Calculations are incorrect or missing. Graphs and tables are missing or incorrect. Results are misinterpreted. (0.5 Marks)
3	Measurements and understanding of concepts (4 Marks)	Observation	Measurements are highly accurate, with minimal deviation from expected values (within $\pm 1\%$). Understanding of concepts is highly	Measurements are accurate, with minor deviations from expected values (within $\pm 5\%$). Understanding of concepts is clear, mostly	Measurements are generally accurate, but with noticeable deviations (within $\pm 10\%$ of expected values). Understanding of concepts is adequate, with some connection	Measurements show significant deviations from expected values (within $\pm 20\%$ of expected values). Understanding of concepts is unclear and poorly connects to objectives.

			insightful, clearly connecting to the objectives, and demonstrates deep understanding. (4 Marks)	connects to the objectives, and shows good understanding. (2 Mark).	s to objectives. (1 Marks)	(0.5 Marks)
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Table-3

Graded Component- 5(15 Marks)

Lab Exam Write-up ,Execution and Viva rubrics (Max: 15 marks)						
Sl. No	Criteria	Measuring Methods	Excellent	Good	Satisfactory	Poor
1.	Datasheet write up (5 Marks)	Visual Aids(Circuit/ logic diagram, Graphs, Tables)	Visual aids are correctly represented, accurate, and well-integrated. (5 Marks)	Visual aids are mostly accurate, with minor issues (3 Mark)	Visual aids are basic, with some inaccuracies or integration issues. (2 Marks)	Visual aids are poorly integrated or often inaccurate (0 Marks)
2.	Execution (4 Marks)	Observations	Circuit matches schematic; correct power/pinouts; instruments set correctly; stable readings. (4 Marks)	Minor corrections needed; readings generally correct. (3 Marks)	Needs guidance; partial correctness; unstable/inconsistent readings. (2 Marks)	Incomplete/unsafe circuit; incorrect or invalid readings (0 Marks)

